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LAB 10: INTER-VLAN ROUTING USING ROUTER-ON-A-STICK

1. OBJECTIVE

To design and implement a virtual local area network (VLAN) topology separating two organizational departments (DEV and HR).

To configure access ports on a Layer 2 switch and assign them to VLAN 10 and VLAN 20.

To establish an 802.1Q trunk connection between the switch and router.

2. NETWORK TOPOLOGY & ADDRESSING SCHEME

All devices are connected using **Copper Straight-Through** cables.

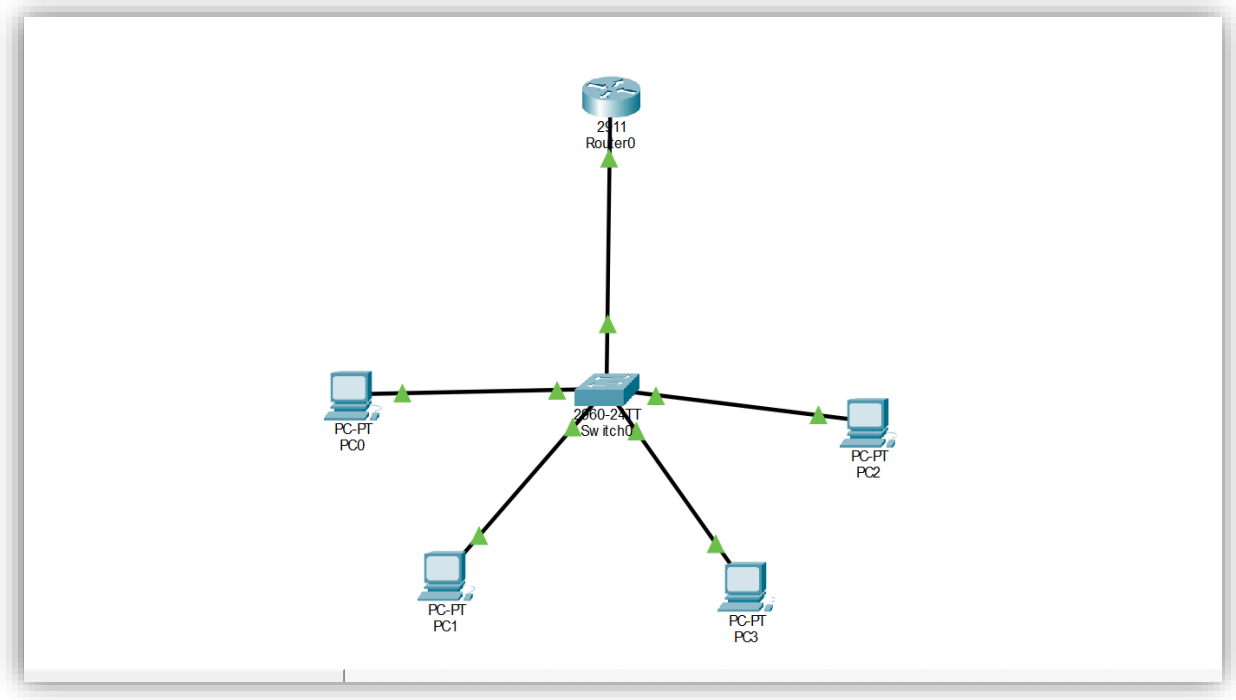


Figure 1: Network Topology for the lab

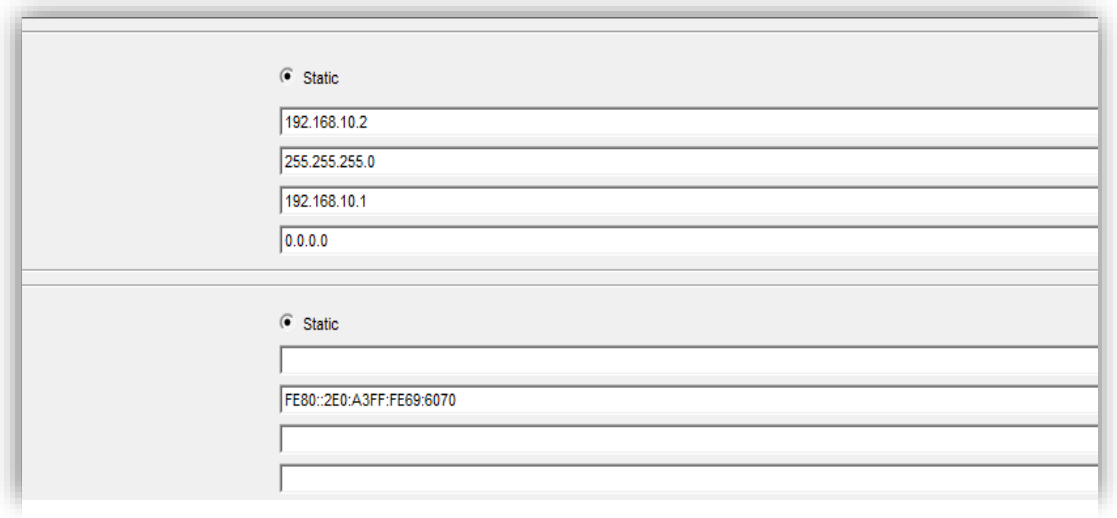


Figure 2: IP Configuration and setting Default Gateways

Device	Connected Port	VLAN / Role	IP Address	Subnet Mask	Default Gateway
PC0	Switch Fa0/1	VLAN 10 (DEV)	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Switch Fa0/2	VLAN 10 (DEV)	192.168.10.3	255.255.255.0	192.168.10.1
PC2	Switch Fa0/3	VLAN 20 (HR)	192.168.20.2	255.255.255.0	192.168.20.1
PC3	Switch Fa0/4	VLAN 20 (HR)	192.168.20.3	255.255.255.0	192.168.20.1
Switch0	Router Gi0/0	Trunk (Fa0/24)	N/A	N/A	N/A
Router0	Subinterface Gi0/0.10	VLAN 10 Gateway	192.168.10.1	255.255.255.0	N/A
Router0	Subinterface Gi0/0.20	VLAN 20 Gateway	192.168.20.1	255.255.255.0	N/A

3. STEP-BY-STEP CONFIGURATION

STEP 1: SWITCH CONFIGURATION (SWITCH0)

CREATE VLAN 10 AND VLAN 20:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)# name DEV
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)# name HR
Switch(config-vlan)#exit
Switch(config)#
```

.

ASSIGN ACCESS PORTS TO VLANS:

```
Switch(config)#
Switch(config)#interface range FastEthernet 0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range FastEthernet 0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

CONFIGURE TRUNK CONNECTION TO ROUTER:

```
Switch(config)#
Switch(config)#interface FastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)#exit
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
```

STEP 2: ROUTER CONFIGURATION (ROUTER0)

ENABLE PHYSICAL INTERFACE:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)# no shutdown

Router(config-if)#exit
```

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CONFIGURE SUB-INTERFACES FOR INTER-VLAN ROUTING:

```
Router(config)#interface GigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface GigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config)#
```

4. TESTING & VERIFICATION

Execute the following ping tests from **PC0** (192.168.10.2) in the Command Prompt:

1. **Intra-VLAN Ping:** ping 192.168.10.3 (PC0 to PC1 in DEV department).
2. **Gateway Ping:** ping 192.168.10.1 (PC0 to Router Sub-interface).
3. **Inter-VLAN Ping:** ping 192.168.20.2 (PC0 in DEV to PC2 in HR department).

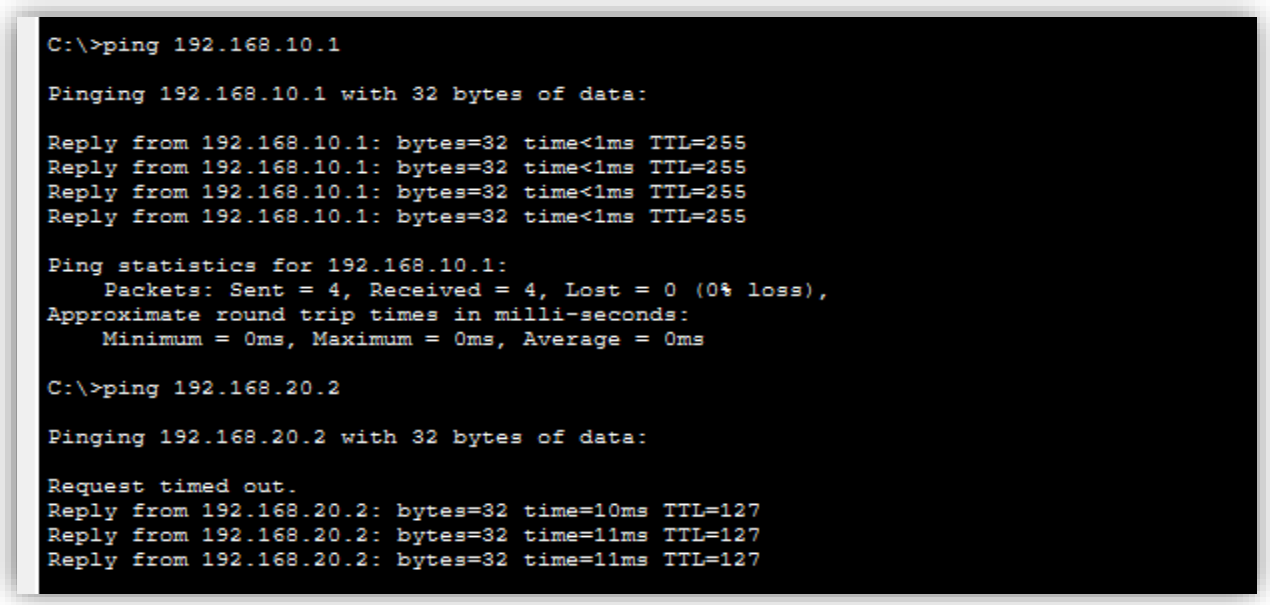


Figure 3: Ping results showing success

5. TEST RESULTS & CONCLUSION

Test Type	Target IP	Executed Command	Result
Same VLAN (DEV)	192.168.10.3	ping 192.168.10.3	**Pass (0% loss)**
VLAN Gateway	192.168.10.1	ping 192.168.10.1	**Pass (0% loss)**
Cross-VLAN (HR)	192.168.20.2	ping 192.168.20.2	**Pass (0% loss)**

CONCLUSION: VLANs 10 and 20 were successfully created and isolated on the switch. Inter-VLAN routing was achieved using the 802.1Q trunk link and router sub-interfaces, enabling seamless cross-department communication.